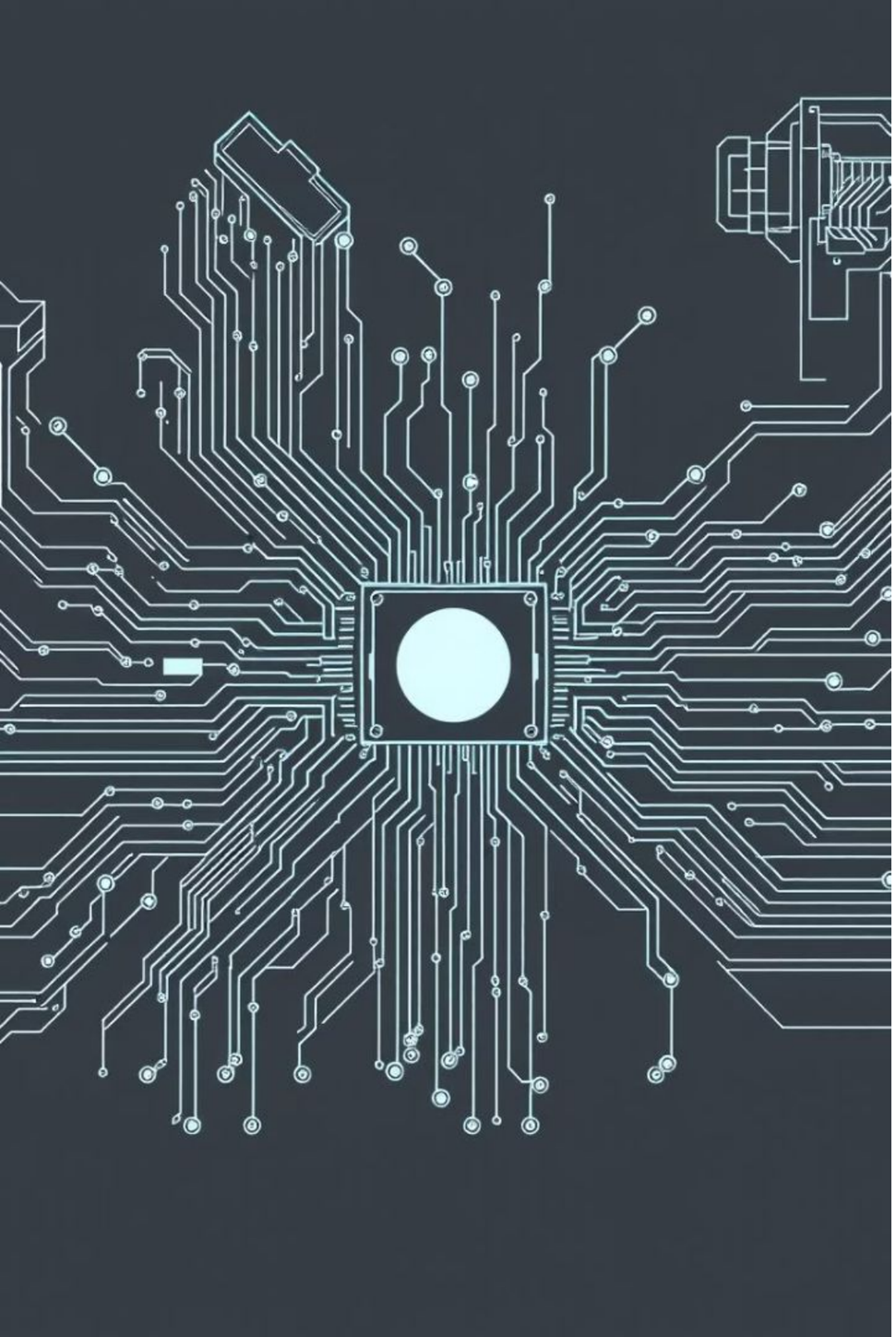

Robotics Project Update: Progress to Date

This presentation reviews our progress on the robotics project.

Made By:

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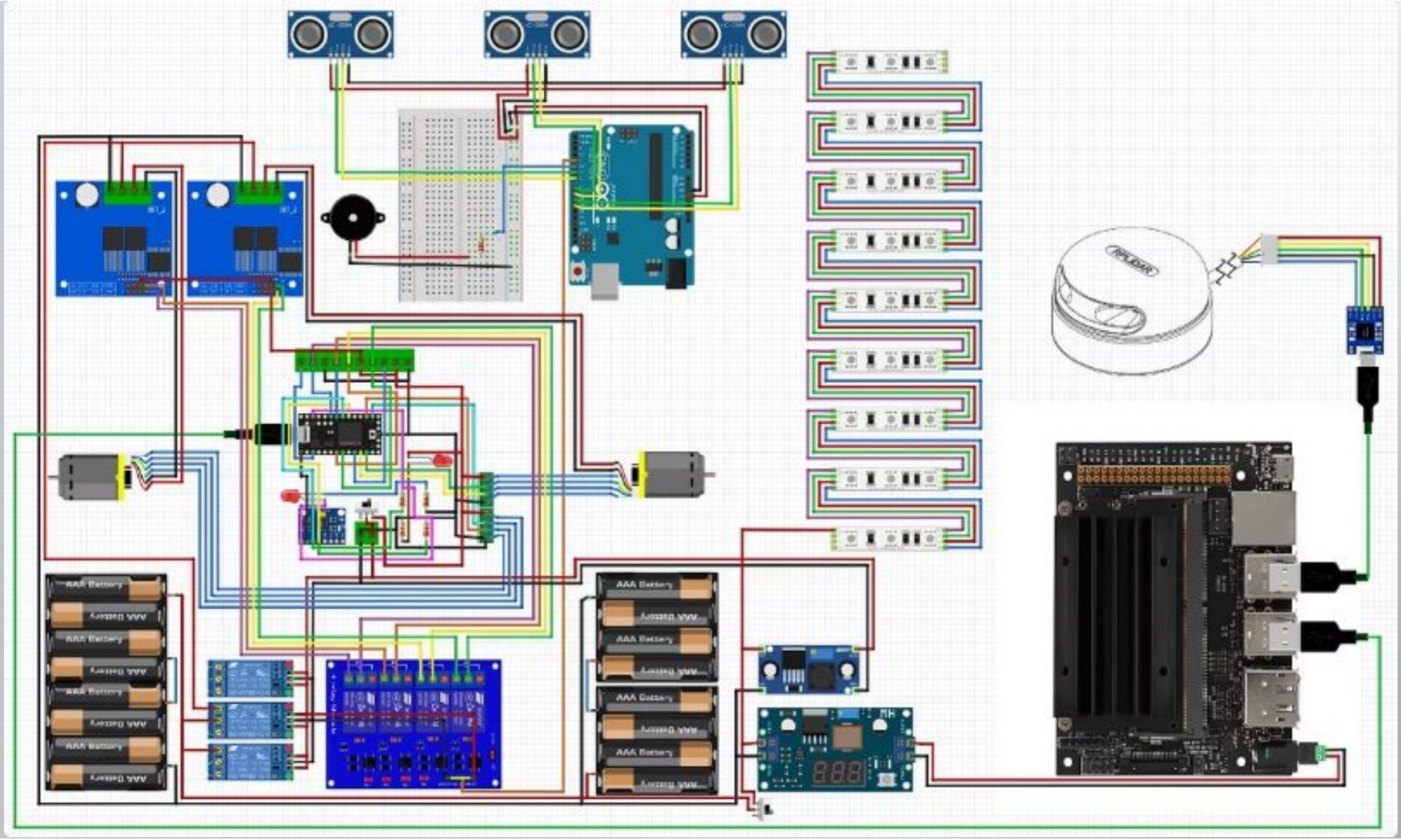
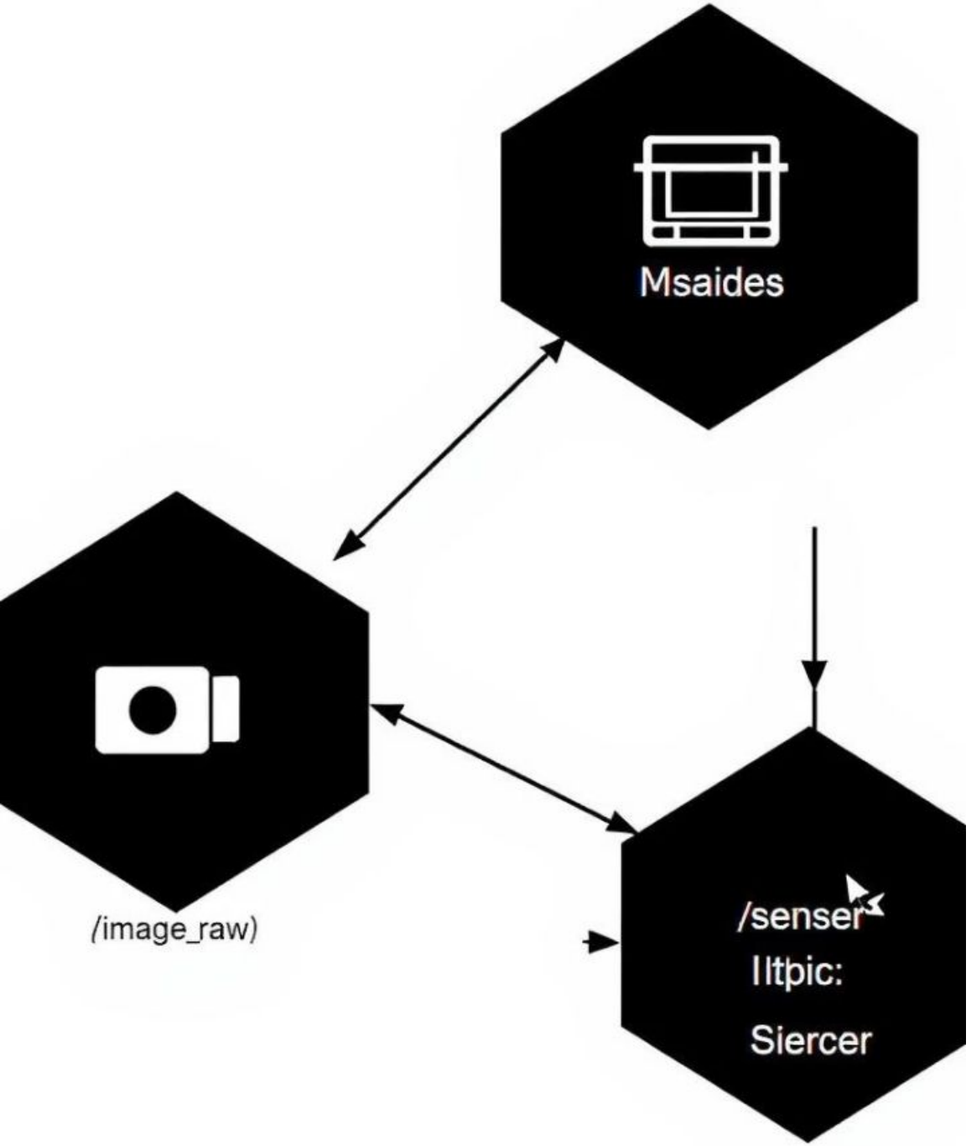


Circuit Diagram Overview



Node and Topic Communication Architecture

ROS Nodes

Modular processes handling sensing, planning, and actuation

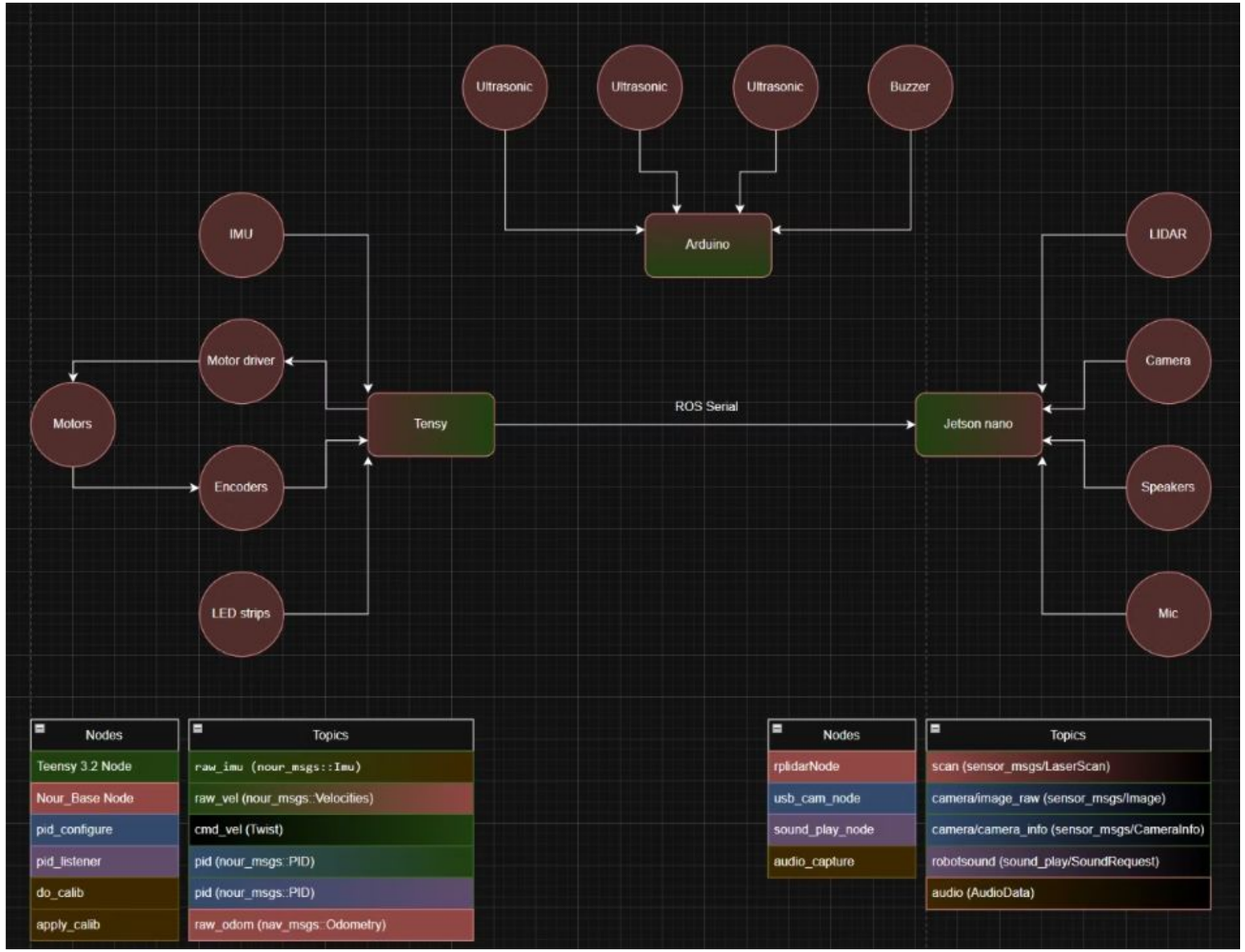
Topics

Defined channels for message exchange between nodes

Data Rates

Optimized for real-time performance and reliability

Node and Topic Communication



Single Button Boot Chart

1 **Power-On**

System starts when the button is pressed

2 **Self-Diagnostics**

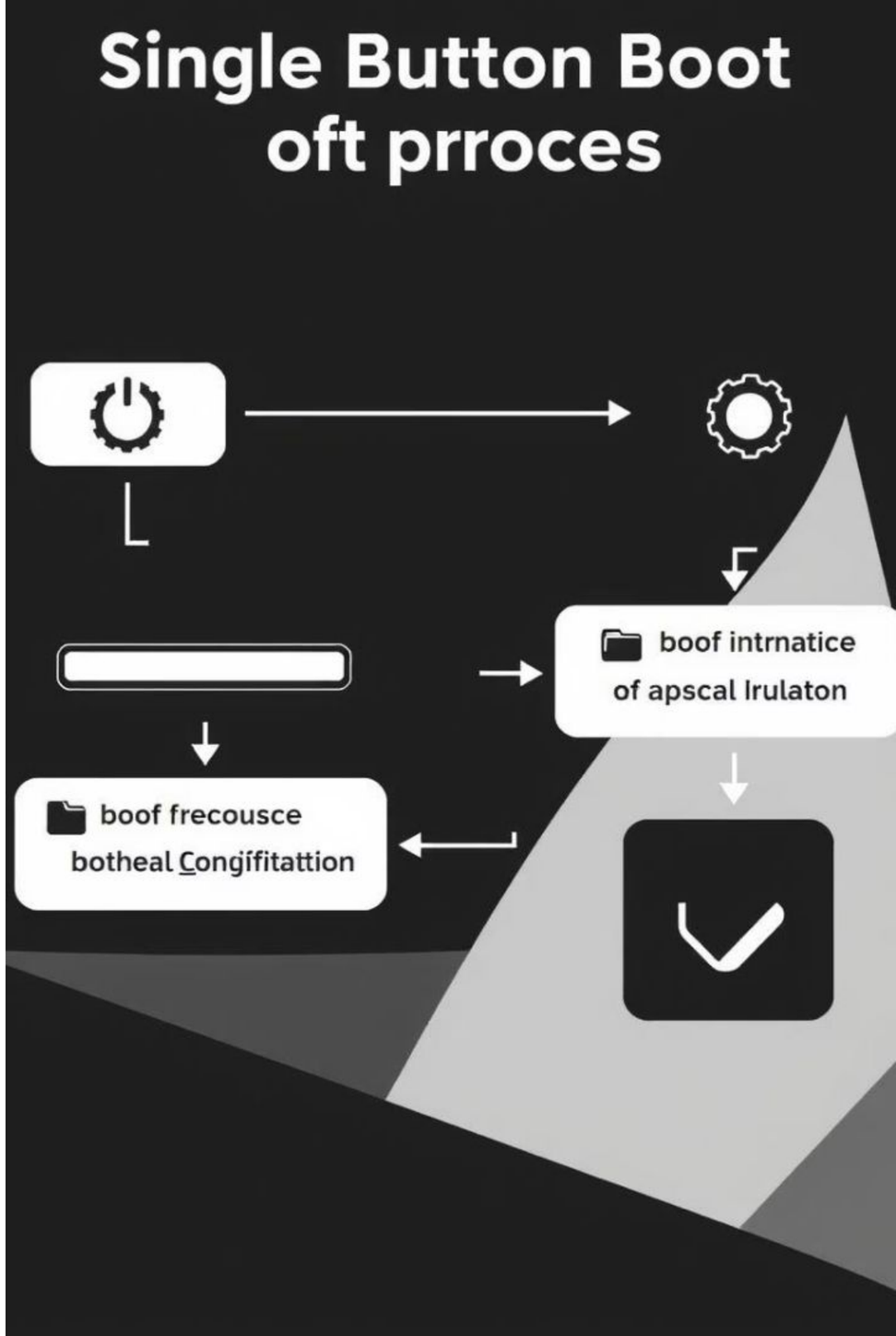
Run hardware and software checks

3 **Initialization**

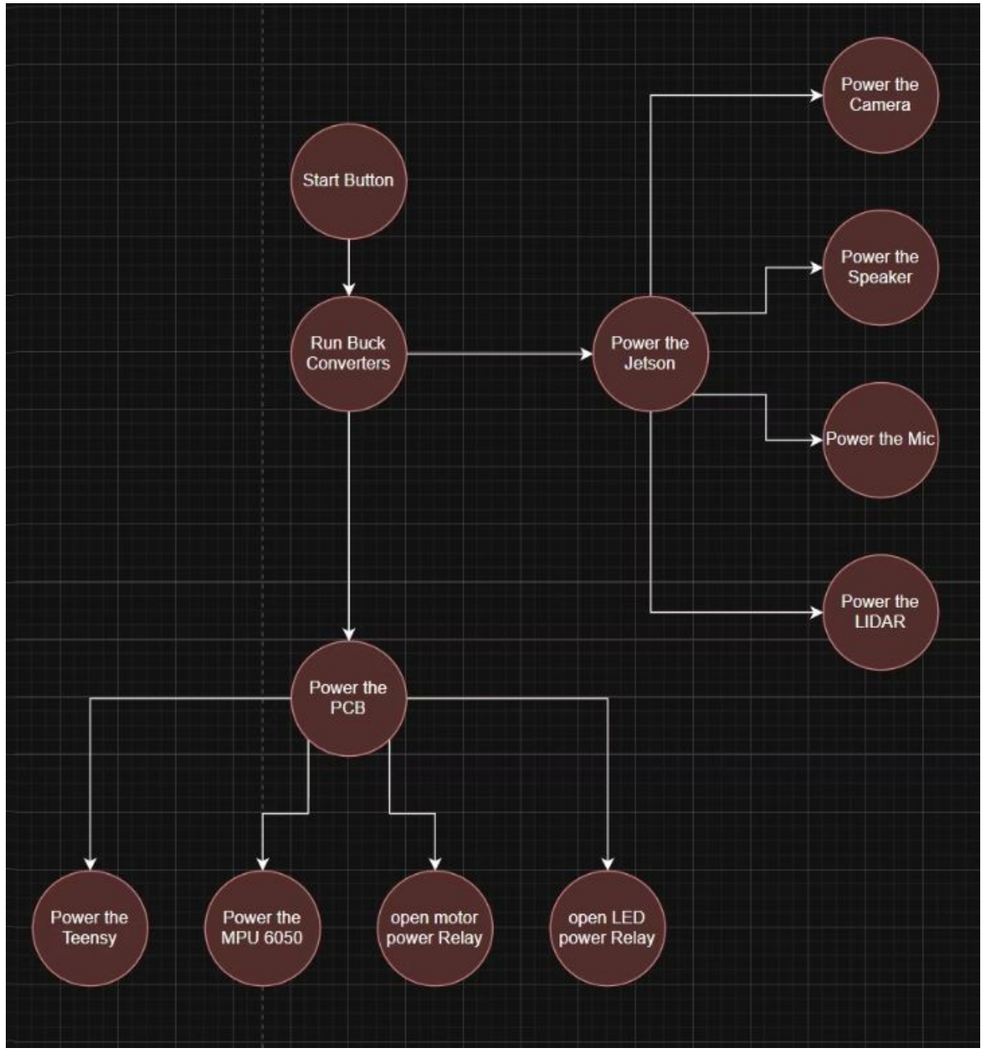
Prepare communication and control modules

4 **Operational State**

Robot ready for tasks



Single Button Boot



Motor Driver Replacement

Original vs New

- Replaced corrupted driver with two new units
- Improved reliability and performance
- Max amphere

Physical Replacement

update the code

Updated pin layout to match dual driver setup

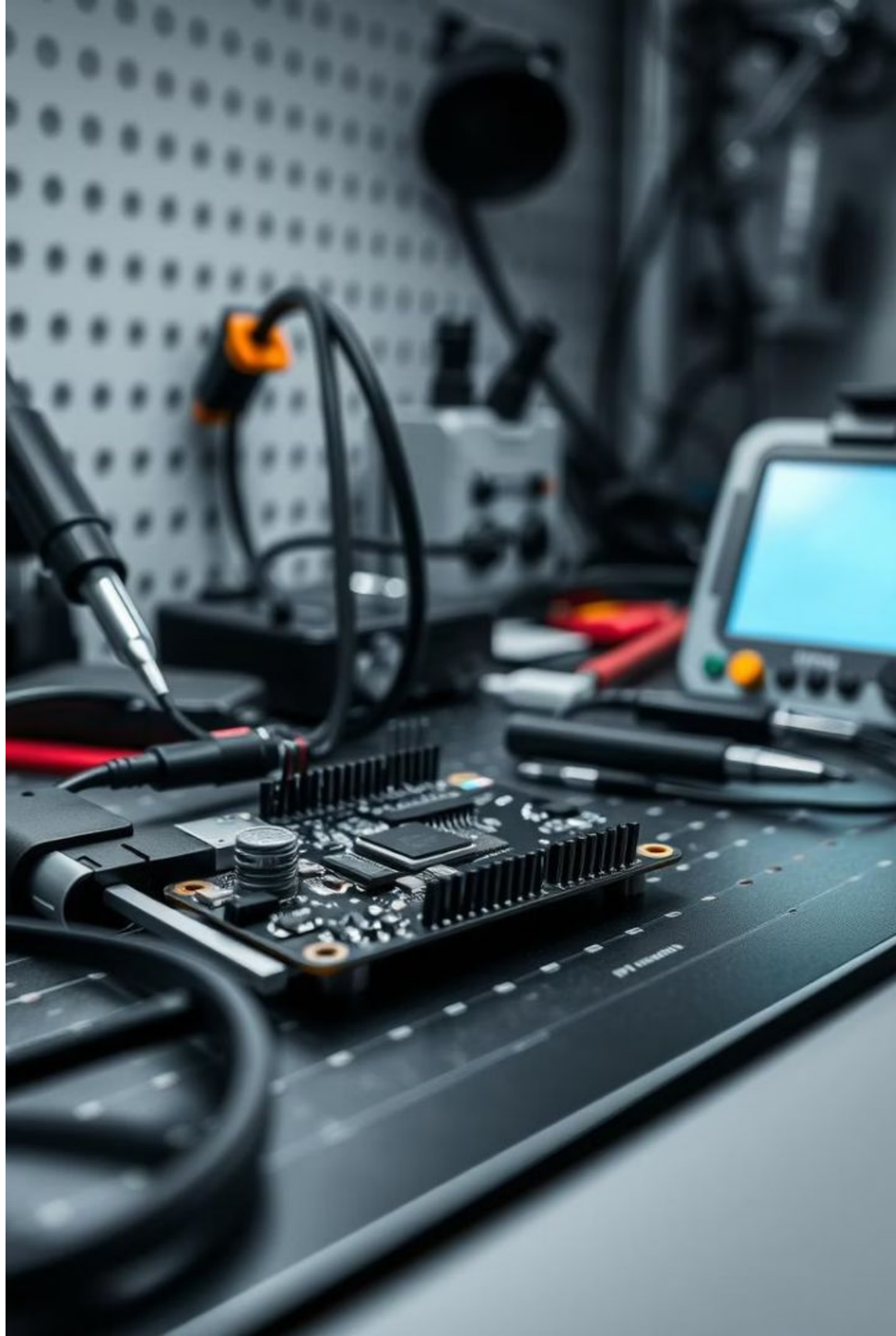
Teensy Repair and Recovery

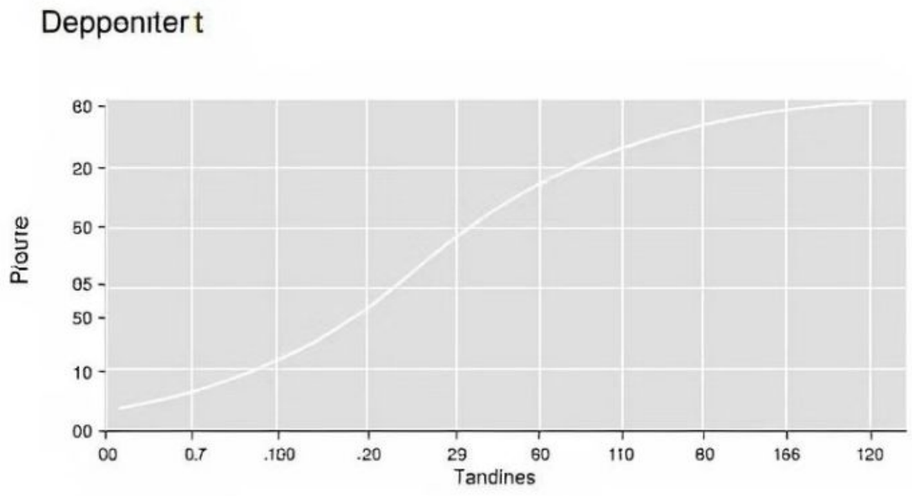
1 **Issue Identification**

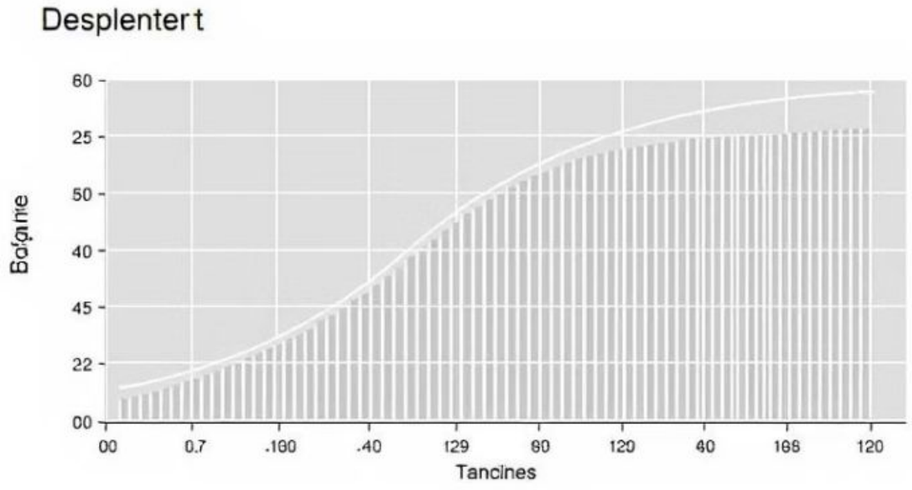
Detected corrupted Teensy firmware causing errors

2 **Troubleshooting Steps**

Used diagnostic tools to isolate problem







PID Constants Optimization

Corrupted Constants

Initial PID values caused unstable motor control

Re-Tuning Process

Systematically adjusted Kp, Ki, and Kd values

Optimized Values

Kp, Ki, Kd improved stability

Project Achievements and Next Steps

Speakers

run the speakers with bluetooth tongle

RealSense Camera

run it and handle its high power demands

microphone to detect sounds

RealSense Camera Addition

1. Handle power spikes if there
2. Testing

Add Microphone

1 Good postioning

2 Testing

Bluetooth Speaker

1. Trying use small Bluetooth module
2. Use Bluetooth Tongle for the jetson

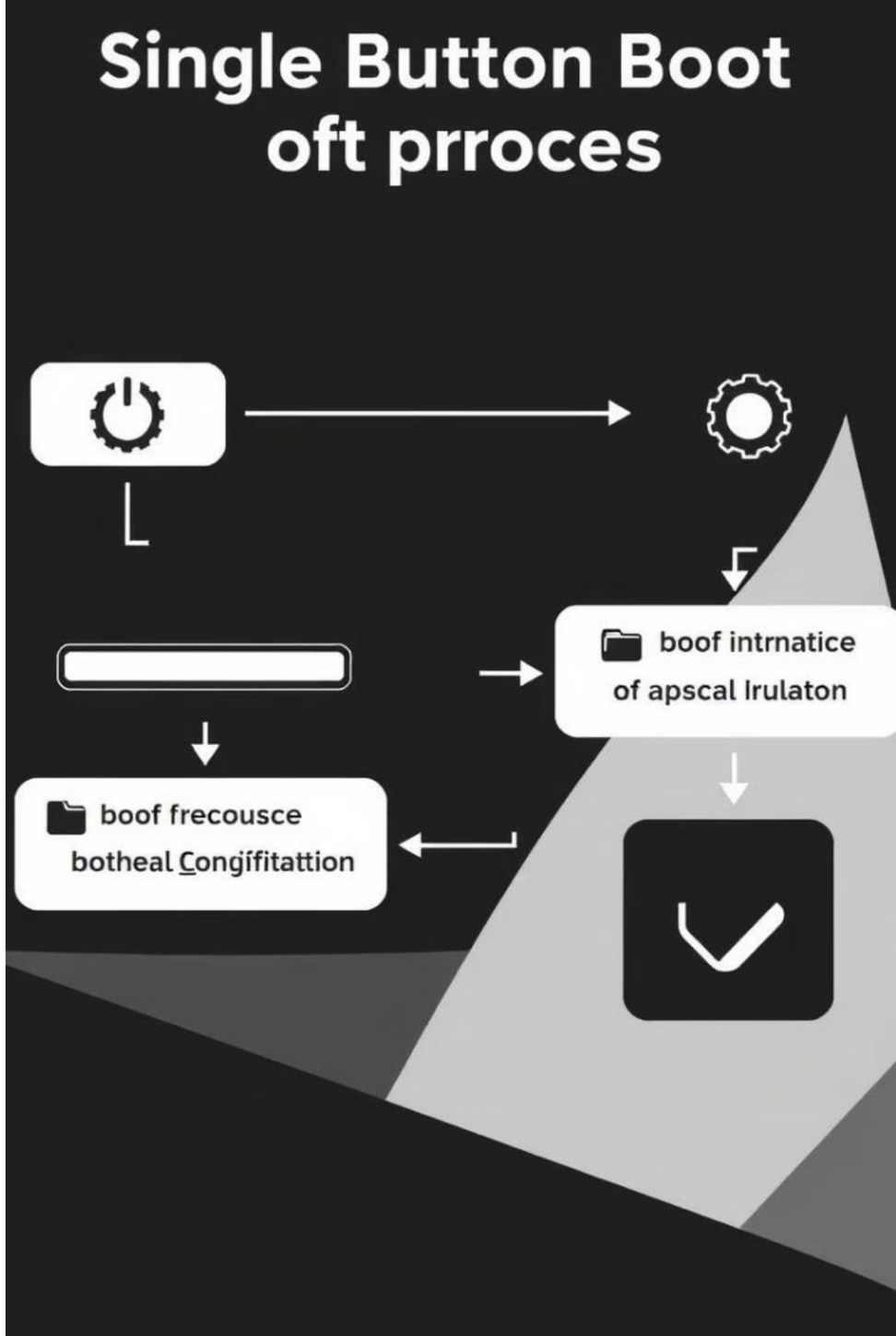
Bluetooth Speaker

3 Make a node to detect cmd_vel topic for any movement

4 testing

Fix the Arduino emergency stop system

1. Replace damaged ultrasonics
2. Fix wiring
3. Power the arduino
4. Testing



Thank YOU